

Water

- At the shampoo bowl, be careful how you handle the spray hose. Position the client's head for comfort and access, being conscious of your own body position as well. Do not bend or twist from the waist unnecessarily. Wipe up any water spills or leaks immediately.
- If the water temperature reaches a scalding level while in the hot position, turn the thermostat on the hot-water tank down to a more acceptable temperature for application to the skin, scalp, and hair. Water heaters should not be set at higher than 130 degrees Fahrenheit.
- As a precaution, always test the water temperature on the inside of your wrist before applying to a client's hair or scalp. The same procedure may be used to test steam towels for facials and shaves.

Electricity

- Electricity and water do not mix. Make sure that all electrical appliances and tools are stored safely, and preferably unplugged, when in proximity to water.
- Never overload electrical outlets or use extension cords.
- Make sure that electrical outlets located near water sources are ground fault circuit interrupter (GFCI) outlets.
- Have shop wiring checked periodically to ensure a safe, fire-free environment.
- Schedule annual safety and operational check-ups for wiring, hot-water tanks, air conditioners, and ventilation systems.

Tools and Appliances

- Tools and equipment should be strategically placed so that the items are safely stored when not in use, yet are accessible when needed. Smaller tools, such as clippers, trimmers, blowdryers, or curling irons, may be placed in countertop receptacles designed for that purpose. Do not forget to disinfect before and after each use. Larger equipment such as UV-light units (used only for storage of disinfected items) may be mounted under the cabinet, attached to a wall, or set on a shelf. Disinfecting jars and electric latherizers are usually placed on the station countertop, but should be set back toward a wall or partition so as not to interfere with other tools. This also limits the risk of accidental burns from a hot latherizer or the spillage of disinfectant solutions.
- If a tool or implement is dropped on the floor during a service, it must be replaced with a disinfected tool or you must stop the service and properly disinfect the tool that was dropped prior to continuing the service. This is a good reason to keep an extra set of tools that are ready to use handy.
- All tools and implements should be in good working condition. Replace damaged tools immediately, including worn electrical cords, chipped clipper blades, cracked housings, and broken shears. Do not try to repair tools yourself; send them to the manufacturer for service. Never subject yourself or your client to the risks of faulty or broken equipment.

- Barbering tools and implements are designed for specific purposes, so use the right tool for the job. Do not expect a trimmer to do the work of a clipper!
- Electrical cords deserve specific mention because they often become a safety hazard in the shop. Cords to clippers, trimmers, curling irons, and blowdryers tend to become twisted and tangled during use. If the cord is too long, it can get caught on the foot or armrests of a hydraulic chair or even on the foot of a client. Some barbers use cordless trimmers to eliminate the problem altogether. A well-planned workstation with sufficient and conveniently placed outlets can also help minimize the *tangled cord syndrome*.
- Never place any tool or implement in your mouth or pocket.

Equipment and Fixtures

- Keep all hydraulic chairs, headrests, shampoo chairs, heat lamps, and lighting fixtures in good working order. Tighten screws and bolts, grease or oil hinges, and service equipment mechanisms as needed.
- Dust and clean regularly to avoid dust buildup and to maintain clean conditions.
- Maintain lighting fixtures. Change bulbs when necessary to keep workstations well lit.

Ventilation

- Proper ventilation and air circulation are extremely important in today's barbershop. Particles from products such as talc, hair sprays, and disinfectants can be inhaled and may cause allergies or other health problems. Heating and air-conditioning vents should be located to perform their optimum functions without interfering with client services. Vents should be vacuumed or cleaned periodically to prevent any buildup of hair that might impede ventilation. Fumes from chemical applications and nail care products require sophisticated filtration units that cleanse and detoxify the air. Once installed, air filters should be changed or cleaned regularly.

Attire

- Clothing should be comfortable and professional in appearance. Excessively baggy clothes can get in the way of your performance just as *easily as tight clothing can restrict it*.
- Long hair worn in a loose style may easily get caught in blowdryer motor vents and other appliances. Keep hair pulled back or short enough to avoid entanglements.
- Necklaces should be of an appropriate length so as not to get caught on equipment or dangle in a client's face at the shampoo bowl or during a shave. Rings should not be worn on the index and middle fingers as they might interfere with haircutting accuracy. In general, rings with stones and elaborate settings are very hard to keep clean. Watches should be waterproof and shock absorbent.

- Shoes should have nonskid rubber soles with good support.
- Electronic devices that may distract you, such as cell phones or tablets, should be kept stored away and only checked or answered between clients.

Children

- Children can cause serious risk of injury to themselves in the shop environment. Being aware of their inquisitive nature and the speed with which they can move can help prevent accidents from happening.
- Post notices in the reception area that advises patrons that children are not to be left unattended.
- Do not allow children to play, climb, or spin on hydraulic chairs.
- Do not allow children to wander freely around the shop with access to workstations, storage areas, and so forth.
- When cutting a child's hair, try to anticipate the child's sudden moves. Never trust a young child to hold his head still while you approach his head area with shears or other tools. Instead, hold the child's head gently but firmly with one hand while cutting with the other. This technique is especially helpful when cutting around ears and at the nape. When trimming bangs or front hairlines, hold the hair between your fingers at a low elevation, cutting the hair on the inside of your palm, thereby putting the barrier of your fingers between the tool and the child's face.

Adult Clients

- As barbers, many of the things we do to assure client comfort also fall under the category of safety precautions. In the following chapters you will learn proper draping procedures and chemical application methods to ensure client safety and comfort from the standpoint of avoiding skin irritations, burns, wet or soiled clothing, and so forth; however, there are also several common-sense services that should be performed. Using good manners and performing common courtesies will help you gain the reputation of being a safety-conscious and courteous barber.
- Assist clients (especially the elderly) in and out of hydraulic and shampoo chairs. Turn hydraulic chairs so the client may get out of the chair without risk of his feet becoming tangled in any of the cords.
- Always lower the hydraulic chair to its lowest level and lock it in position so that it does not spin before inviting the client to be seated or leave the chair.
- Hold doors open for clients.
- Assist clients in walking whenever necessary.
- Always support the back of the chair, and thus the client, when reclining or raising a chair back.
- Cushion the client's neck with a folded towel in the neck rest at the shampoo bowl. This is especially beneficial for clients with neck injuries such as whiplash.

- Support the client's head whenever appropriate at the shampoo bowl. Do not ask the client to hold his head up for shampooing or rinsing. Instead, gently turn the client's head to the side for easier access to the back and nape areas.

Exits

- Exits should be well marked and identifiable. (Check with your local building inspection office for codes and requirements.)
- Employees should know where exits are located and how to evacuate the building quickly in case of fire or other emergencies. Implement fire drills to practice for this contingency.

Fire Extinguishers

- Fire extinguishers should be placed where they are readily accessible.
- All employees should be instructed in fire extinguisher use.
- It is a law that fire extinguishers be checked periodically. Be guided by the manufacturer's recommendations and state and local ordinances.

Chemicals

- Request an SDS from your supplier(s) for each product purchased for use or sale in the shop, and maintain an SDS notebook. Add to the notebook as new products are brought into the workplace.
- Take the time to study the SDS from the manufacturer so that you will know how and where to store products and what procedures to follow in an emergency.
- There is a correct way to dispose of chemicals such as haircolor tints, chemical relaxers, and bleach. Contact your local hazardous-waste department or agency for disposal guidelines.
- Never mix cleaning products together under any circumstances.
- Never mix leftover chemicals together.
- Do not allow leftover chemicals to stockpile. Dispose of products as often as necessary to maintain a safe environment.
- Check the inventory stock occasionally for plastic bottles with dents or bulges. This swelling of the container indicates that the contents are under pressure and could possibly explode.
- Wear goggles when pouring or mixing products. Add a lab apron and gloves when handling any caustic or skin-irritating material.
- Chemical spills require an absorbent material, such as sawdust, to clean up properly. Be guided by your local hazardous-waste agency.
- Every solution, liquid, cream, powder, paste, gel, and other substance should be properly labeled. This requirement goes beyond labeled products purchased from a supplier and includes spray bottles of water or setting lotions, disinfecting jars, containers of blade wash, and any other substance used in the shop that is not contained within its original packaging.

List Your Professional Responsibilities

After reading this section, you will be able to:

LO8 List your responsibilities as a professional barber.

After studying this chapter, it should be clear that your responsibilities as a professional barber far exceed the ability to perform a good haircut; your most important responsibility is to protect your clients' health and safety.

- Never take shortcuts for cleaning and disinfecting. You cannot afford to skip steps or save money when it comes to safety.
- It is your professional and legal responsibility to follow state and federal laws and rules.
- Keep your license current and notify the licensing agency if you move or change your name.
- Check your state's website monthly for any change or update to the rules and regulations.
- Be aware of your environment so that you can identify and eliminate potential hazards to make the barbershop safer for you and your clients.
- Be prepared for emergencies. Every shop should have employee and clientele emergency information available near the telephone.
 - An emergency phone number checklist should include the contact numbers for fire, police, and medical rescue departments; the nearest hospital emergency room; and taxis.
 - Utility service companies, such as electricity, water, heat, air-conditioning, and landlord or custodial numbers are also helpful in an emergency or if something breaks down in the shop. Update this information on an annual basis and you will always be prepared.
- Realize that behavior that stems from a knowledgeable and caring manner is what separates a true professional from a nonprofessional, and being a professional is something you can take pride in.

CLEANING AND DISINFECTING: NONELECTRICAL TOOLS AND IMPLEMENTS

Nonelectrical tools and implements include combs, brushes, shears, clips, hairpins, tweezers, and all other items that can be completely immersed into a disinfectant container.

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Covered storage container
- ☐ Disinfectant
- ☐ Disposable gloves
- ☐ Disposable towels
- ☐ Liquid disinfectant
- ☐ Liquid soap
- ☐ Safety glasses
- ☐ Scrub brush
- ☐ Timer
- ☐ Tongs

PROCEDURE



1 It is important to wear safety glasses and gloves while disinfecting nonelectrical tools and implements to protect your eyes from unintentional splashes of disinfectant, to prevent possible contamination of the implements by your hands, and to protect your hands from the powerful chemicals in the disinfectant solution.



2 Prepare the cleaning solution according to the manufacturer's directions.

3 Remove all visible hair from your tools and implements before washing.



- 4 Rinse all implements with warm running water.



- 5 Use a small scrubbing brush to clean them in the cleaning or soap solution.



- 6 Brush grooved items thoroughly and open hinged implements to scrub the revealed areas clean.



- 7 Rinse away all traces of solution or soap with warm running water. The presence of soap in most disinfectants will cause them to become inactive. Soap is most easily rinsed off in warm, not hot, water.



- 8 Dry implements thoroughly with a clean or disposable towel, or allow them to air dry on a clean towel.

- 9 Your implements are now properly cleaned and ready to be disinfected.



10 Immerse cleaned implements in an appropriate disinfection container holding an EPA-registered disinfectant for the required time (at least 10 minutes or according to the manufacturer's instructions). Remember to open hinged implements before immersing them in the disinfectant. If the disinfectant solution is visibly dirty, or if the solution has been contaminated, it must be replaced.



11 After the required disinfection time has passed, remove tools and implements from the disinfectant solution with tongs or gloved hands, rinse the tools and implements well in warm running water, and pat them dry.



12 Store dry, disinfected tools and implements in a clean, covered container until needed.



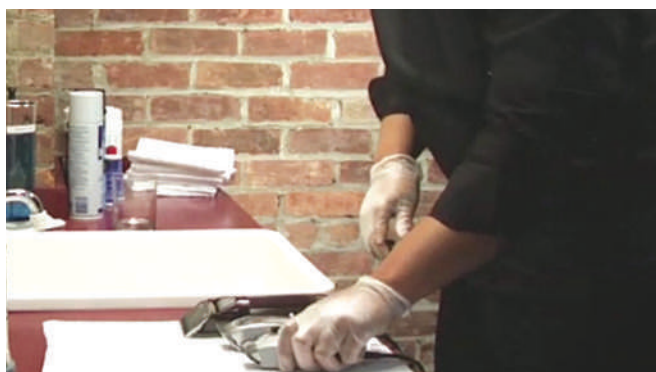
13 Remove gloves and thoroughly wash your hands with warm running water and liquid soap. Rinse and dry hands with a clean fabric or disposable towel.

CLEANING AND DISINFECTING: CLIPPERS AND OUTLINERS

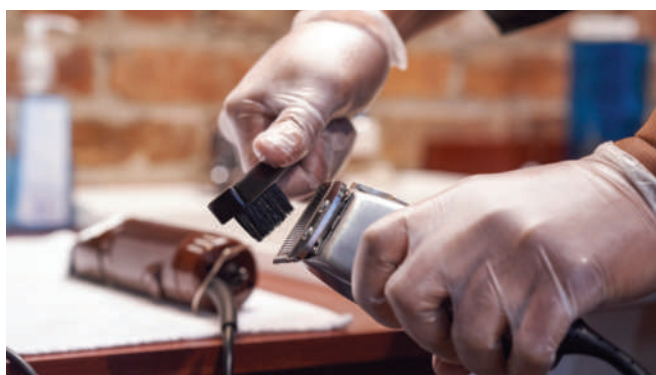
The disinfection of electrical tools, such as clippers and outliners, requires a different approach than that for nonelectric tools and implements. Hair particles and bacteria become trapped between and behind clipper blades; so thorough cleaning and disinfection of these tools is very important. There are a large number of disinfectant choices that work well for clipper blades. Again, it is vital that regardless of what type you choose, the label must be EPA registered as bactericidal, virucidal, and fungicidal, and in some states, even as tuberculocidal. Many of the products that have been used historically do not meet the current requirements of most states, so it is important to know what those requirements are and to follow them, regardless of historical practice.

Setup for Cleaning and Disinfecting Clippers and Outliners

PROCEDURE



1 Arrange all supplies, products, and tools on a clean surface.



2 Remove hair particles from clipper blades with a stiff brush.

Cleaning and Disinfecting Clippers and Outliners with Nondetachable Blades

3 To clean the blades and prepare them for disinfection, you must follow one of the following two methods.

CLEANING WITH BLADE WASH



- 4 Pour blade wash into a glass, plastic, or disposable container wide enough to accommodate the width of the clipper blades to a depth of approximately $\frac{1}{2}$ inch.



- 5A For nondetachable blades, submerge only the cutting teeth of the clipper blades into the blade wash and turn the unit on. Run the blades in the solution until no hair particles are seen being dislodged from between the blades.



- 5B For detachable clipper blades, detach the blades and submerge into the blade wash for the manufacturer's recommended time.



- 6 Wipe the blades dry with a clean towel.

CLEANING WITH AEROSOL CLEANER



7 Use aerosol spray cleaner as indicated on the label, following the manufacturer's instructions for use. Spray the cleaner until you can see the blades getting wet.



8 Turn the unit on and let the blades run until you see hair being dislodged from the blades.

9 Wipe the blades dry with a clean, dry towel.

Disinfecting Using a Disinfectant Spray



10 Now that your clippers are clean, spray with a clipper disinfectant and allow blades to remain moist for full contact time as mentioned on the manufacturer's label.

Grease or Oil Clippers and Outliners

11 For all of your clippers or outliners, it is important to grease or oil the parts regularly to keep your tools in top shape.



12 To oil the blades, put one drop of clipper oil on each side of the blades.



13 Next, turn the unit on and let it run for 15 to 30 seconds.

14 Turn off the clippers or outliners and unplug.

CLEANING AND DISINFECTING CONDUCTOR CORDS



15 Before you begin cleaning the conductor cord, always double-check that it is unplugged.

16 To clean the conductor cord, put soap on a wet towel. Lather up the towel and run the soapy towel down the cord.



17 Using a wet towel, rinse off the cord by running the towel down the cord.



18 With a dry towel, dry the cord completely.



19 Then, wipe down the cord with a disinfectant wipe, ensuring that cord is never dropped to the floor or off of the clean working surface.

20 Store cleaned and disinfected clippers, or outliners, in a clean, covered container until needed for use.

21 Make sure to use one of these methods after each client.

PROPER HAND WASHING

Hand washing is one of the most important procedures in your infection control efforts and is required in every state before beginning any service.

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- Disposable paper towels
- Liquid soap in a pump container
- Nail brush

PROCEDURE



1 Turn the water on, wet your hands.



2 Pump soap from a pump container onto the palm of your hand.



3 Rub your hands together, all over and vigorously, until a lather forms. Continue for a minimum of 20 seconds.



4 Choose a clean and disinfected nail brush.



5 Wet the nail brush, and pump soap on to the bristles.



6 Brush your nails horizontally back and forth under the free edges.



7 Change the direction of the brush to vertical and move the brush up and down along the nail folds of the fingernails. The process for brushing both hands should take about 60 seconds to finish.



8 Rinse hands in running warm water.



9 Use a clean cloth or paper towel for drying your hands, according to the barbershop's policies or state rules and regulations.



10 After drying your hands, turn off the water with the towel. Use the towel to open the door and then dispose of the towel. Touching a doorknob with your bare fingers can recontaminate your hands.

HANDLING AN EXPOSURE INCIDENT

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Antiseptic
- ☐ Bandages
- ☐ Cotton
- ☐ Disposable gloves
- ☐ Disposable paper towels
- ☐ Liquid soap
- ☐ Plastic bag

PROCEDURE

Should you accidentally cut a client, calmly take the following steps:



1 Stop the service immediately.



2 Dispose of the blade in a sharps container and place the razor in a container designated for cleaning and disinfection.



3 Face your client and calmly apologize for the incident.



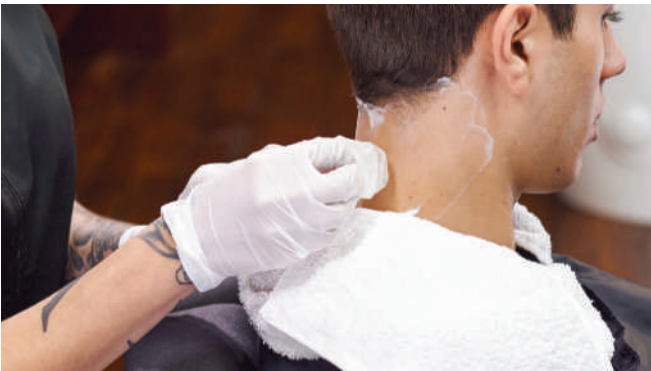
4 Excuse yourself to go wash your hands.



- 5 Once your hands are clean, immediately put on gloves.
 - 6 Apply slight pressure to the area with a moistened cotton round to stop the bleeding.
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- 7 Dispose of the used cotton round in a plastic storage bag.
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- 8 Then gently clean with an antiseptic and have the client wash the area if appropriate.
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- 9 Dispose of the antiseptic wipe in the plastic storage bag.
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- 10 Dispense styptic powder onto a cotton round and use a cotton swab to apply it to the injury to stop any residual bleeding.



- 11 Apply an adhesive bandage to completely cover the wound.



12 Clean and disinfect the workstation, as necessary.



13 Discard all single-use, contaminated objects such as wipes or cotton balls in a plastic bag and then place in a trash bag. Deposit sharp disposables in a sharps box. Dispose of double-bagged items and sharps containers as required by state or local law. Information on these laws may be found on your local board website or through the OSHA website. In general, all of these items (except sharps) may go into the regular trash.



14 Remember, before you remove your gloves, all tools and implements that have come into contact with blood or other body fluids must be thoroughly cleaned and completely immersed in an EPA-registered hospital disinfectant solution designed for 10 minutes. Blood may carry pathogens, so you should never touch an open sore or a wound. Gloves should be removed with caution to avoid touching the contaminated surface of the glove. Remove one glove by holding it in other gloved hand, sliding your finger under the glove at the base of the glove, and pulling it over the other contaminated glove—never allowing either glove to come into contact with your skin.

15 Wash your hands with soap and warm water before returning to the service. Recommend that the client see a physician if any signs of redness, swelling, pain, or irritation develop.

REVIEW QUESTIONS

- 1 What is the primary purpose of regulatory agencies?
- 2 What is an SDS? Where can they be obtained?
- 3 List the four types of microorganisms that are pertinent to a barber.
- 4 What is a contagious disease?
- 5 Is HIV a risk in the barbershop? Why or why not?
- 6 What is the difference between cleaning, disinfecting, and sterilizing?
- 7 What is complete immersion?
- 8 List at least nine safety tips to follow when using disinfectants.
- 9 What are the characteristics of items that can be disinfected?
- 10 What does porous mean? Name two porous items that can be cleaned and disinfected in between use on clients.
- 11 What are Standard Precautions?
- 12 What is an exposure incident?
- 13 Describe the procedure for handling an exposure incident on a client in the barbershop.
- 14 List the four things that must be on a disinfectant label for approved use in all states.

CHAPTER GLOSSARY

acquired immune deficiency syndrome (uh-KWY-erd ih-MYOON di-FISH-en-see sin-droh-m)	p. 70	abbreviated AIDS; a disease that breaks down the body's immune system; AIDS is caused by the human immunodeficiency virus (HIV)
acquired immunity (uh-KWY-erd ih-MYOO-nih-tee)	p. 71	immunity that the body develops after overcoming a disease, through inoculation (such as flu vaccinations), or through exposure to natural allergens (such as pollen, cat dander, and ragweed)
allergy (AL-ur-jee)	p. 73	reaction due to extreme sensitivity to certain foods, chemicals, or other normally harmless substances
antiseptics (ant-ih-SEP-tiks)	p. 79	chemical germicide formulated for use on skin; registered and regulated by the Food and Drug Administration
asymptomatic (A-symp-toe-mat-ic)	p. 80	showing no symptoms or signs of infection
bacilli (bah-SIL-ee)	p. 65	short rod-shaped bacteria; they are the most common bacteria and produce diseases such as tetanus (lockjaw), typhoid fever, tuberculosis, and diphtheria
bacteria (bak-TEER-ee-ah)	p. 63	single-celled microorganisms that have both plant and animal characteristics; some bacteria are harmful; some are harmless
bacterial spores (bak-TEER-ee-ul SPOORZ)	p. 63	bacteria capable of producing a protective coating that allows them to withstand very harsh environments and to shed the coating when conditions become more favorable to them
bactericidal (bak-TEER-uh-syd-uhl)	p. 63	capable of destroying bacteria
binary fission (BY-nayr-ee FISH-un)	p. 66	the division of bacterial cells into two new cells called <i>daughter cells</i>
biofilms	p. 69	colonies of microorganisms that adhere to environmental surfaces, as well as the human body

bloodborne pathogens (BLUD-boorn PATH-o-genz)	p. 69	disease-causing microorganisms carried in the body by blood or body fluids, such as hepatitis and HIV
cilia (SIL-ee-uh)	p. 65	hairlike extensions that protrude from cells and help sweep away fluids and particles
cleaning (KLEEN-ing)	p. 63	a mechanical process (scrubbing) using soap and water or detergent and water to remove all visible dirt, debris, and many disease-causing germs; cleaning also removes invisible debris that interferes with disinfection; cleaning is what barbers are required to do before disinfecting
cocci (KOK-sy)	p. 65	round-shaped bacteria that appear singly (alone) or in groups; the three types of cocci are staphylococci, streptococci, and diplococci
communicable (kuh-MYOO-nih-kuh-bul)	p. 67	able to be communicated; transferable by contact from one person to another as in a communicable disease
contagious disease (kun-TAY-jus DIZ-eez)	p. 67	also known as <i>communicable disease</i> ; disease that is spread from one person to another person; some of the more contagious diseases are the common cold, ringworm, conjunctivitis (pinkeye), viral infections, and natural nail or toe and foot infections
contamination (kun-tam-uh-NAY-shun)	p. 68	the presence, or the reasonably anticipated presence, of blood or other potentially infectious materials on an item's surface or visible debris or residues such as dust, hair, and skin
diagnosis (dy-ag-NOH-sis)	p. 68	determination of the nature of a disease from its symptoms and/or diagnostic tests; federal regulations prohibit salon professionals from performing a diagnosis
diplococci (dip-loh-KOK-sy)	p. 65	spherical bacteria that grow in pairs and cause diseases such as pneumonia
direct transmission (dy-REKT trans-MISH-uhn)	p. 64	transmission of blood or body fluids through touching (including shaking hands), kissing, coughing, sneezing, and talking.
disease (DIZ-eez)	p. 60	an abnormal condition of all or part of the body, or its systems or organs, that makes the body incapable of carrying on normal function
disinfectants (dis-in-FEK-tents)	p. 60	chemical products approved by the EPA designed to destroy most bacteria (excluding spores), fungi, and viruses on surfaces
disinfection (disinfecting) (dis-in-FEK-shun)	p. 63	a chemical process that uses specific products to destroy harmful organisms (except bacterial spores) on environmental surfaces
efficacy (ef-ih-KUH-see)	p. 74	the ability of a product to produce the intended effect; on a disinfectant label, it indicates specific pathogens destroyed or disabled when used properly
exposure incident (eks-POH-zhoor in-SI-dent)	p. 80	contact with non intact (broken) skin, blood, body fluid, or other potentially infectious materials, which is the result of the performance of an employee's duties
flagella (fluh-JEL-uh)	p. 65	slender, hairlike extensions used by bacilli and spirilla for locomotion (moving about); may also be referred to as <i>cilia</i>
fungi (FUN-ji)	p. 70	single-celled organisms that grow in irregular masses and include molds, mildews, and yeasts; they can produce contagious diseases such as ringworm
fungicidal (FUN-ji-syd-uhl)	p. 63	capable of destroying molds and fungi
hepatitis (hep-uh-TY-tus)	p. 69	a bloodborne virus that causes disease and can damage the liver
hospital disinfectants (HOS-pih-tal dis-in-FEK-tents)	p. 60	disinfectants that are effective for cleaning blood and body fluids on nonporous surfaces
human immunodeficiency virus (HYOO-mun ih-MYOO-noh-di-FISH-en-see VY-rus)	p. 70	abbreviated HIV; virus that causes HIV disease and acquired immune deficiency syndrome (AIDS)

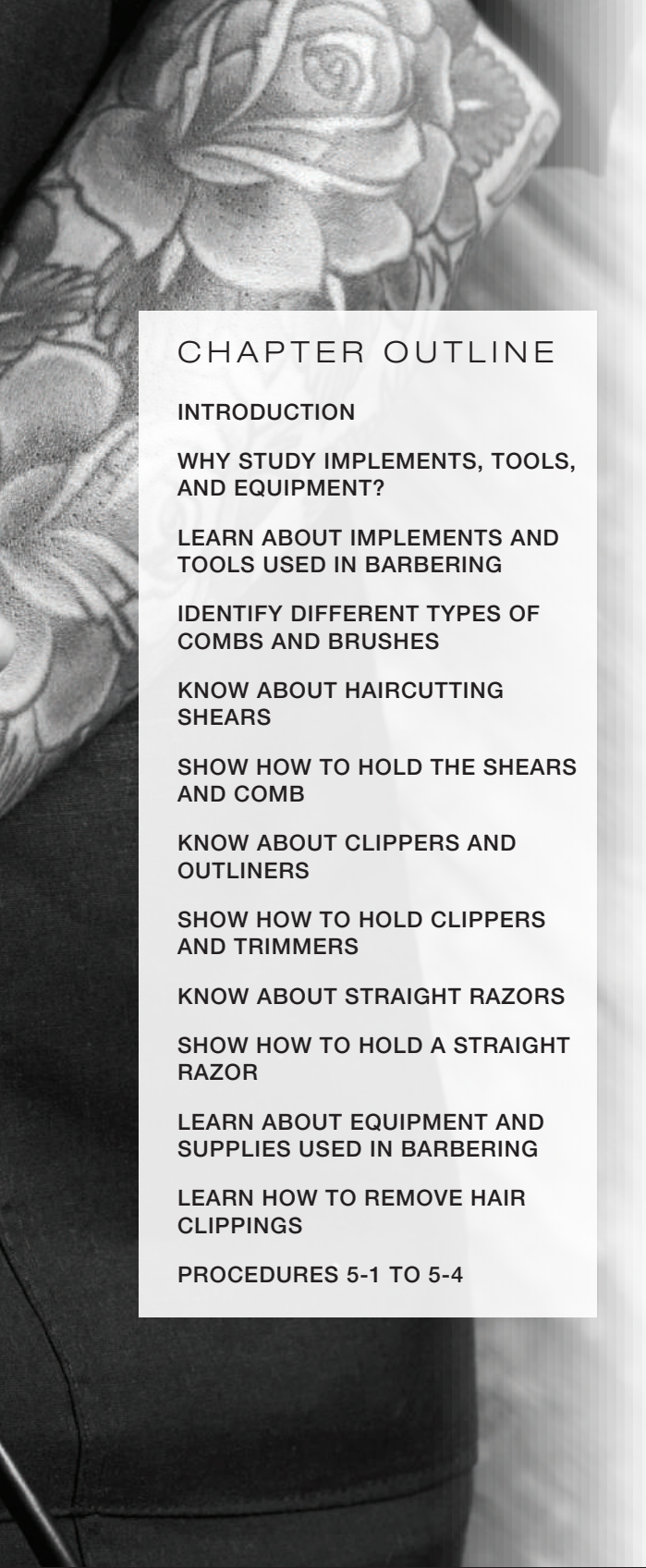
immunity (im-YOO-net-ee)	p. 71	the ability of the body to destroy and resist infection; immunity against disease can be either natural or acquired and is a sign of good health
Inactive stage (IN-ac-tif STAJE)	p. 66	also known as <i>spore-forming stage</i> ; the ability of some bacteria to form a protective coating around themselves to protect them from harsh environments during the inactive stage
indirect transmission (IN-dih-rekt trans-MISH-uhn)	p. 64	transmission of blood or body fluids through contact with an intermediate contaminated object such as a razor, extractor, nipper, or an environmental surface
infection (in-FEK-shun)	p. 62	the invasion of body tissues by disease-causing pathogens
infection control (in-FEK-shun CON-trol)	p. 62	the methods used to eliminate or reduce the transmission of infectious organisms
infectious (in-FEK-shus)	p. 62	caused by or capable of being transmitted by infection
infectious disease (in-FEK-shus DIZ-eez)	p. 63	disease caused by pathogenic (harmful) microorganisms that enter the body; an infectious disease may or may not be spread from one person to another person
inflammation (in-fluh-MAY-shun)	p. 66	a condition in which the body reacts to injury, irritation, or infection, characterized by redness, heat, pain, and swelling
local infection (LOHK-al in-FEK-shun)	p. 67	an infection, such as a pimple or abscess, that is confined to a particular part of the body and appears as a lesion containing pus
methicillin-resistant staphylococcus aureus (METH-eh-sill-en-ree-ZIST-ent staf-uh-loh-KOK-us OR-ee-us)	p. 67	abbreviated MRSA; a type of infectious bacteria that is highly resistant to conventional treatments such as antibiotics
microorganism (my-kroh-OR-gah-niz-um)	p. 63	any organism of microscopic or submicroscopic size
mildew (MIL-doo)	p. 70	a type of fungus that affects plants or grows on inanimate objects but does not cause human infections in the barbershop
motility (MOH-till-it-ee)	p. 65	self-movement
multiuse (mul-tih-YUS)	p. 73	also known as <i>reusable</i> ; items that can be cleaned, disinfected, and used on more than one person, even if the item is accidentally exposed to blood or body fluid
natural immunity (NATCH-uh-rul im-YOO-net-ee)	p. 71	immunity that is partly inherited and partly developed through healthy living
nonpathogenic (nahn-path-uh-JEN-ik)	p. 64	harmless microorganisms that may perform useful functions and are safe to come in contact with since they do not cause disease or harm
nonporous (nahn-POHW-rus)	p. 60	an item that is made or constructed of a material that has no pores or openings and cannot absorb liquids
occupational disease	p. 68	illness resulting from conditions associated with employment, such as prolonged and repeated overexposure to certain products or ingredients
parasites (PAYR-uh-sytz)	p. 71	organisms that grow, feed, and shelter on or inside another organism (referred to as the <i>host</i>), while contributing nothing to the survival of that organism. Parasites must have a host to survive
parasitic disease	p. 68	disease caused by parasites, such as lice and mites
pathogenic (path-uh-JEN-ik)	p. 64	harmful microorganisms that can cause disease or infection in humans when they invade the body
pathogenic disease	p. 68	disease produced by organisms, including bacteria, viruses, fungi, and parasites

pediculosis capitis (puh-dik-yuh-LOH-sis KAP-ih-tus)	p. 71	infestation of the hair and scalp with head lice
phenolic disinfectants (fi-NOH-lik dis-in-FEK-tents)	p. 75	tuberculocidal disinfectants that are a form of formaldehyde, have a very high pH, and can damage the skin and eyes
porous (POHW-rus)	p. 78	made or constructed of a material that has pores or openings; porous items are absorbent
pus (PUS)	p. 67	a fluid created by infection
quaternary ammonium compounds (KWAT-ur-nayr-ee uh-MOH-nee-um KAHM-powndz)	p. 75	commonly known as quats are products made of quaternary ammonium cations and are designed for disinfection of nonporous surfaces; they are appropriate for use in noncritical (noninvasive) environments and are effective against most pathogens of concern in the barbershop environment
Safety Data Sheet (SAYF-tee DAY-tuh SHEET)	p. 58	abbreviated SDS; required by law for all products sold. SDSs include safety information about products compiled by the manufacturer, including hazardous ingredients, safe use and handling procedures, proper disposal guidelines, and precautions to reduce the risk of accidental harm or overexposure
sanitation (san-ih-TAY-shun)	p. 56	also known as <i>sanitizing</i> ; a chemical process for reducing the number of disease-causing germs on cleaned surfaces to a safe level
scabies (SKAY-beez)	p. 71	a contagious skin disease that is caused by the itch mite, which burrows under the skin
sodium hypochlorite (soh-DEE-um hy-puh-KLOR-yt)	p. 75	common household bleach; an effective disinfectant for the barbershop
spirilla (spy-RIL-ah)	p. 65	spiral or corkscrew-shaped bacteria that cause diseases such as syphilis and Lyme
Standard Precautions (STAN-dard PRUH-caw-shuns)	p. 80	are guidelines published by the CDC that require the employer and employee to assume that any human blood and body fluids are potentially infectious
staphylococci (staf-uh-loh-KOKS-sy)	p. 65	pus-forming bacteria that grow in clusters like a bunch of grapes. They cause abscesses, pustules, and boils
sterilization (stayr-ih-luh-ZAY-shun)	p. 72	the process that completely destroys all microbial life, including spores
streptococci (strep-toh-KOK-sy)	p. 65	pus-forming bacteria arranged in curved lines resembling a string of beads. They cause infections such as strep throat and blood poisoning
tinea barbae (TIN-ee-uh BAR-bee)	p. 70	also known as <i>barber's itch</i> , a superficial fungal infection that commonly affects the skin; it is primarily limited to the bearded areas of the face and neck or around the scalp
tinea capitis (TIN-ee-uh KAP-ih-tus)	p. 70	a fungal infection of the scalp characterized by red papules, or spots, at the opening of the hair follicles
toxins	p. 71	various poisonous substances produced by some microorganisms (bacteria and viruses)
tuberculocidal disinfectants (tuh-bur-kyoo-LOH-syd-ahl dis-in-FEK-tents)	p. 60	often referred to as phenolics, are proven to kill the bacterium that cause tuberculosis, in addition to other pathogens destroyed through the use of hospital disinfectants
tuberculosis (tuh-bur-kyoo-LOH-sus)	p. 60	a disease caused by bacteria that are transmitted through coughing or sneezing
virucidal (VI-ra-syd-uhl)	p. 63	capable of destroying viruses
virus (VY-rus)	p. 67	a parasitic submicroscopic particle that infects and resides in cells of biological organisms. A virus is capable of replication only through taking over the host cell's reproductive function

5



IMPLEMENTS, TOOLS, AND EQUIPMENT



CHAPTER OUTLINE

INTRODUCTION

WHY STUDY IMPLEMENTS, TOOLS,
AND EQUIPMENT?

LEARN ABOUT IMPLEMENTS AND
TOOLS USED IN BARBERING

IDENTIFY DIFFERENT TYPES OF
COMBS AND BRUSHES

KNOW ABOUT HAIRCUTTING
SHEARS

SHOW HOW TO HOLD THE SHEARS
AND COMB

KNOW ABOUT CLIPPERS AND
OUTLINERS

SHOW HOW TO HOLD CLIPPERS
AND TRIMMERS

KNOW ABOUT STRAIGHT RAZORS

SHOW HOW TO HOLD A STRAIGHT
RAZOR

LEARN ABOUT EQUIPMENT AND
SUPPLIES USED IN BARBERING

LEARN HOW TO REMOVE HAIR
CLIPPINGS

PROCEDURES 5-1 TO 5-4

LEARNING OBJECTIVES

After completing this chapter, you will be able to:

LO1

List the principal tools of the trade used in barbering.

LO2

Describe when to use different combs and brushes.

LO3

Discuss and identify the types of haircutting shears.

LO4

Identify the parts of haircutting shears.

LO5

Show how to properly hold shears for haircutting.

LO6

Show how to palm the shears and comb.

LO7

Describe two types of clippers.

LO8

Identify the main parts of a clipper.

LO9

Show different ways to hold clippers for haircutting.

LO10

Name two types of straight razors.

LO11

Identify the different parts of a straight razor.

LO12

Show how to hold a straight razor for shaving, honing,
and stropping.

LO13

Show how to hold a straight razor for haircutting.

LO¹⁴

Describe the functions of hones and strops.

LO¹⁵

Show how to hone and strop a conventional blade straight razor.

LO¹⁶

Identify the types of equipment and supplies used in barbering.

LO¹⁷

Identify ways to remove hair clippings.

LO¹⁸

Show how to perform two towel-wrapping methods.

Introduction

The practice of barbering requires a variety of implements, tools, and equipment. You should strive to purchase high-quality items that will perform effectively and safely to get the job done. This does not mean that you have to purchase the most expensive items, but beware of “bargains.” For example, it is unlikely that a comb that can be bent between your thumb and index finger will actually comb through the hair. Low-quality implements, tools, and equipment are a waste of time and money and can end up costing more to replace than what you would have spent in the first place. When taken care of properly, well-made implements and tools will provide years of dependable service. Since the variety of choices can be confusing, ask your instructor or an experienced barber to assist you when selecting and purchasing new implements, tools, or equipment.

why study

IMPLEMENTS, TOOLS, AND EQUIPMENT?

Barbers should study and have a thorough understanding of implements, tools, and equipment because:

- Understanding the purpose and capabilities of your implements and tools will help you select the right tool for the job.
- Knowing the parts of your implements, tools, and equipment will help you understand instructions given during practical work, to comply with infection control procedures, and to know when an item may need repair or replacement.
- Understanding how to hold and use your implements and tools is critical to achieving desired results.
- Using barbering equipment correctly and safely is important to your safety and the comfort and safety of your clients.

Learn about Implements and Tools Used in Barbering

After reading this section, you will be able to:

LO1 List the principal tools of the trade used in barbering.

As a barber, your principal *tools of the trade* are combs, brushes, shears, clippers, outliners, and razors. In addition, you will use appliances such as blowdryers and thermal styling tools to perform finishing and styling work on your clients. Choosing the right implement or tool for the job will depend on your understanding of the item's functions and characteristics and how to use it in an effective and safe manner.

Identify Different Types of Combs and Brushes

After reading this section, you will be able to:

LO2 Describe when to use different combs and brushes.

KNOW YOUR COMB BASICS

Combs are made from various materials in different shapes and sizes. A good quality comb is just slightly flexible with smooth, rounded teeth that will comb through the hair without snagging individual strands. Most of the combs used in haircutting measure between 7 and 8 inches long with a shape that may be straight, tapered, or handled. The materials used in the construction of the comb determine its characteristics or qualities:

- Hard rubber combs are slightly flexible and durable but can deteriorate if left in a disinfectant over an extended time.
- Combs made from carbon materials are antistatic, stiff, and heat and chemical resistant.
- Graphite combs are less rigid than carbon material combs and are heat and chemical resistant.
- Metal combs can retain heat to help set-in a hairstyle. Made from tempered aluminum or other metals, metal combs may corrode in reaction to chemicals, such as hydrogen peroxide.

DID YOU KNOW?
According to Ace, a Newell Rubbermaid Company, the first hard rubber comb was made in 1851.

The Comb Teeth

The spacing between the teeth of a comb varies with the style of comb. Many combs are designed with fine or narrow (close together) teeth at one end and coarse or wide (farther apart) teeth at the other end. Other comb styles have uniformly spaced teeth for the entire length of the comb.

figure 5-1
Assorted all-purpose combs.

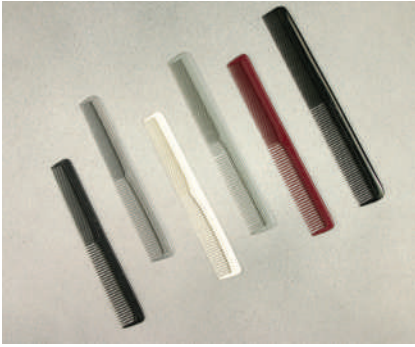


figure 5-2
Taper combs.

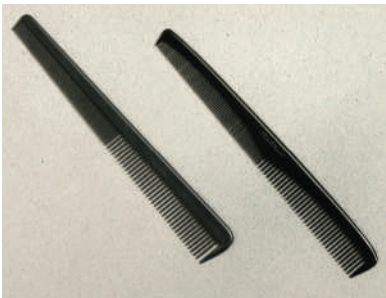


figure 5-3
Wide-tooth combs.



DID YOU KNOW? Using a light-colored comb on dark hair or a dark-colored comb on light hair provides greater contrast between the comb and the hair, which can help you simplify and perfect your work.

The spacing of the teeth is important because the thickness and distribution of the hair strands that flow through the comb can affect your cutting or styling outcomes. For example, combing through a section of hair with the narrow end of the comb merges the hair strands closer together than the wide end of the comb does; thus, the narrow end allows for better control of the hair and more precise haircutting.

Generally speaking, combination combs give you the ability to adapt to different densities within the hair without switching combs. Narrow-toothed combs can be useful for combing and cutting fine hair textures or when performing detail work. Wide-toothed combs are preferable for detangling, creating styling effects, or for distributing product through the hair.

Comb Styles

Given all the different types of combs available, the most appropriate or best comb to use ultimately depends on the procedure to be performed, the client's hair texture, and the barber's personal preference.

- A styling or all-purpose comb, usually constructed of both narrow and wide-spaced teeth, is used for general haircutting and styling (**Figure 5-1**).
- A **taper comb** (TAY-pur KOM) can be used for cutting hair in those areas where a gradual blending of the hair is required. The tapered end is especially useful when working in tight areas or for trimming mustaches, for tapering necklines, or for blending around the ears (**Figure 5-2**).
- A flat handle comb with evenly spaced teeth works best to achieve a flat-top style and is also an option to use in clipper-over-comb cutting.
- Wide-toothed combs can be used to control larger amounts of hair, spread relaxer cream, detangle hair, or comb through curly hair textures (**Figure 5-3**).
- The tail comb is the best choice for sectioning longer hair or when making partings to create subsections, for wrapping the hair on rods or rollers, or for color applications (**Figure 5-4**).
- A hair pick is an effective choice for combing through textured or tightly curled hair (**Figure 5-5**).

figure 5-4
Tail combs.



figure 5-5
Pick combs.



figure 5-6
Proper comb holding position.



figure 5-7
Improper comb holding position.

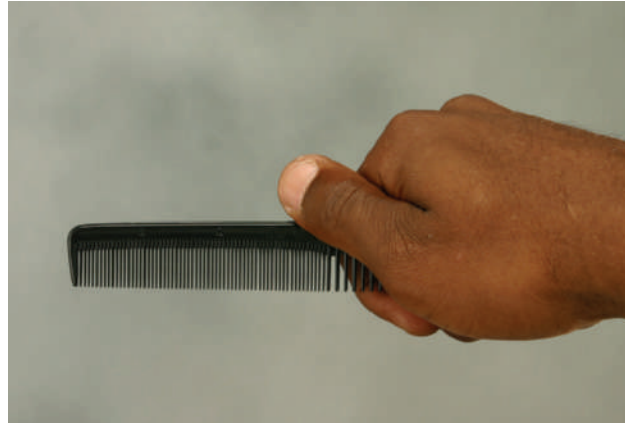


figure 5-8
Proper shear-over-comb position.



figure 5-9
Improper shear-over-comb position.



Holding the Comb

The correct manner in which to hold the comb will be dictated by the type of comb used, the service being performed, and the dexterity and comfort of the barber. **Figures 5-6** through **5-9** show correct and incorrect holding positions with a styling comb. Be guided by your instructor and practice, practice, practice!

KNOW YOUR BASIC BRUSHES

Styling brushes are used to control, smooth, wave, or add fullness to hair or to stimulate the scalp. The choice of bristle texture, spacing, and material will depend on the hair texture, the hairstyle to be achieved, and the barber's personal preference. Most hairbrushes are manufactured with plastic, rubber, wood, or metal bases with natural, synthetic, combined, or metal bristles (**Figure 5-10**). Hairbrushes are available in a variety of shapes and sizes for different purposes (**Table 5-1**).

figure 5-10
Assorted brushes.



table 5-1
TYPES OF BRUSHES

Type of Brush	Characteristics	Use(s)
Synthetic bristle brush	Flexible or rigid nylon bristles set into a cushioned, plastic, or cushioned base.	General brushing and detangling.
Natural bristle brush	Typically made of boar bristles, set into a cushioned or rigid base; traps particles and polishes the hair by distributing sebum through the strands.	General brushing and detangling on fine, delicate, and relaxed hair types.
Mixed bristle brush	Combination of boar and synthetic bristles set into a cushioned or rigid base.	General brushing and detangling.
Paddle brush	Available with all bristle and base types in a rectangular or oval shape.	General brushing and detangling.
Round brush	Available with all bristle types in metal or plastic bases in different diameters.	Creates volume and/or curl in blowdry styling.
Vented brush	Rigid base has openings or “vents” to facilitate quicker drying.	General drying and styling work.
Wet brush	Stiff bristles designed to avoid snagging wet hair.	Detangling wet hair.

Know about Haircutting Shears

After reading this section, you will be able to:

- LO 3 Discuss and identify the types of haircutting shears.
- LO 4 Identify the parts of haircutting shears.

Your haircutting shears are one of the most important tools you will use when performing barbering services. Therefore, your understanding of shears and your ability to select the right type, style, and size that will fit you comfortably is vital to your future success as a barber.

RECOGNIZE SHEAR STYLES

There are two basic styles of shears generally used by barbers: the French style with a finger rest or tang for the little finger and the German style without a tang. Barbers often choose the French-style shear because the tang helps to ensure balance and control during cutting (Figure 5-11).

Cast versus Forged Shears

Professional haircutting shears are either cast or forged during manufacture.

Cast shears are made by heating steel to a liquid form for pouring into a mold. As the metal cools, it expands; this process produces a more porous and weaker steel that does not retain its sharpness, is more brittle, and is likely to chip or break. Because cast shears are less expensive to produce than forged shears, they are also usually less expensive to purchase.

figure 5-11
Haircutting shears.



Forged shears (FORJed SHEERZ) are made by working heated metal into a finished shape through the processes of hammering or compression. In today's shear manufacturing, the metal is either dropped on to a mold to be pounded into the desired shape or is hydraulically pressed into a die mold. The heating and cooling processes are repeated until the desired density, hardness, and shape of the implement has been achieved.

A forged shear may be constructed in one or two pieces and is more durable, stays sharp longer, and is of higher quality than a cast shear.



DID YOU KNOW?

It is very difficult to determine whether shears are forged or cast by sight alone. Be sure to research the manufacture of the shears before making a purchase and do not rely on a salesperson to have all the answers you will need.

KNOW THE BASIC PARTS OF HAIRCUTTING SHEARS

Haircutting shears are composed of two blades called the moving blade and the still blade that are fastened with a tension screw or adjustment knob. Other basic parts of the shears include the points (tips), cutting edges, shanks, bumper, finger grip, thumb grip, and finger rest (tang) (Figure 5-12). Let's take a closer look at these features and other important information you need to know.

- Shear blades.** The cutting edge is the part of the blade that does the cutting. The *grind* refers to the inside construction of the blade and the way it is cut in preparation for sharpening and polishing. The flat grind is traditionally associated with beveled shears, while hollow-ground blades are typically a feature of convex shears. There are also shear designs with replaceable blades that eliminate the need for sharpening.
- Blade edges.** The cutting edges may be convex, beveled, or semi-convex in design (see Table 5-2 and Figure 5-13).

figure 5-12
Parts of haircutting shears.

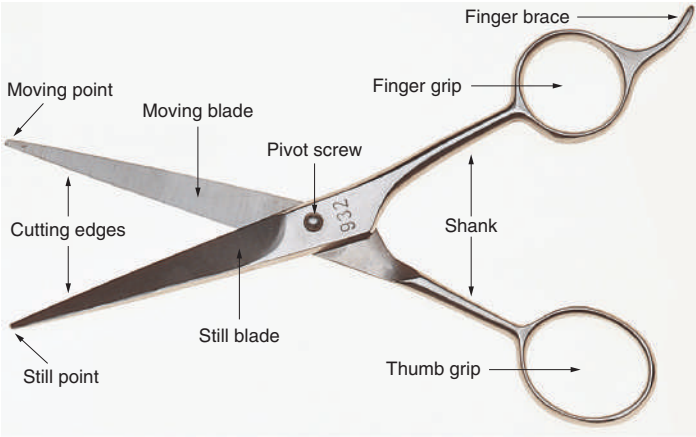
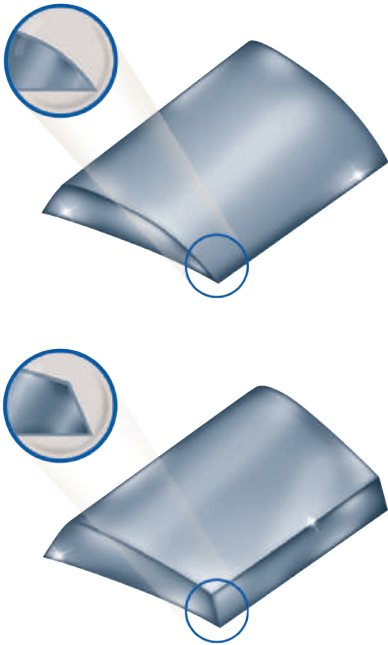


table 5-2
COMPARE BLADE EDGES

Convex Edge	Beveled Edge
Very sharp edges	Usually has serrated edges
Glides through hair easily	Holds hair on blade edge; reduces slippage
Cuts hair shaft cleanly	Grips hair shaft for cutting
Less tension used when cutting	More tension used when cutting
Good for dry, wet, and slide cutting	Not intended for slide cutting
Reduced noise while cutting	Serrations create some noise while cutting
More subject to nicks than beveled blade	Durable; stays sharp longer than convex blade
Sharpen every 4 to 6 months	Sharpen every 9 to 12 months

figure 5-13
Convex and beveled edges.



- The outside of a *convex* blade edge has a clamshell or half-moon shape and the inside is concave or hollow-ground. This arrangement produces a smoother and sharper cutting blade that requires more frequent sharpening than beveled edges.
- A *beveled* edge means there is an angle on the cutting surface of the blade. Beveled edges can be plain-ground with smooth, polished, or razor-edged surfaces. Beveled shears usually have one plain and one corrugated (serrated) blade. The design of the corrugations helps to hold the hair in place and to prevent it from sliding on the blade while it is being cut. A beveled edge will stay sharper longer than a convex blade but tends to require more tension when cutting hair.
- A semi-convex blade consists of a convex blade that has a bevel ground onto the blade edge. This design reduces production costs and extends the time in-between sharpening.
- **Set.** The **set of the shears** refers to the angle of the blade from its tip to the ride and the alignment of the blades in relation to each other. This alignment is just as important as the grind of the blades because even shears with the finest cutting edges will be inferior cutting tools if the blades are not set properly. Tension also affects the set and alignment of the blades and should be checked frequently.
- **Tension screw.** The tension screw or knob, also called a pivot screw, controls the distance between the blades. Refer to the Maintain Your Shears section later in this chapter to learn how to perform an adjustment.
- **Shank.** The *shank* is located between the ride and the base of the finger grip. When the ring finger is positioned properly in the finger grip, the index and middle fingers will rest on the shank.
- **Finger grip and tang.** The *finger grip* or *ring* is the part of the still blade of the shears where you will position your ring finger between the first and second knuckle. To use your shears properly and to maintain control, do not use your middle finger or allow the finger grip to slide past the second knuckle of your ring finger. The *finger tang* is where your little finger can rest for greater control of the shears and to relieve pressure on your nerves and tendons.
- **Thumb grip.** The *thumb grip* or *ring* is a part of the moving blade of the shears. Your thumb will control the moving blade and should rest no lower than your cuticle within the grip.
- **Bumper.** The *bumper* (not pictured in Figure 5-12) is usually positioned on the thumb grip to prevent the grips from touching when the shears are completely closed.
- **Size.** Shears are available in a variety of lengths and are measured in inches and half inches. Convex shears are measured from the tip (points) to the end of the finger ring where it joins the finger rest (Figure 5-14). Beveled shears are measured from the tip to the end of the finger rest. Most barbers prefer shears that are 6½" to 7½" in length because they facilitate shear-over-comb cutting better than shorter shears (Figure 5-15).

DID YOU KNOW?
You can reduce the diameter of a finger or thumb grip for a more comfortable fit by inserting a shear ring or guard into the grip.

figure 5-14

Convex shears are measured from the tip to the end of the finger ring.

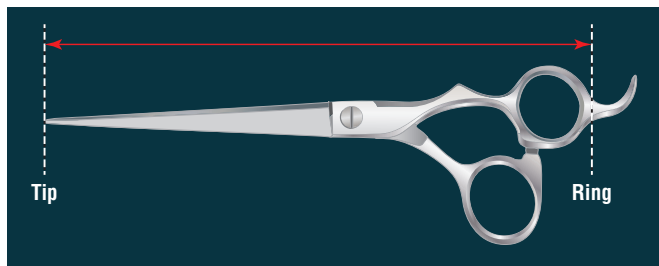


figure 5-15

Beveled shears are measured from the tip to the end of the finger rest.

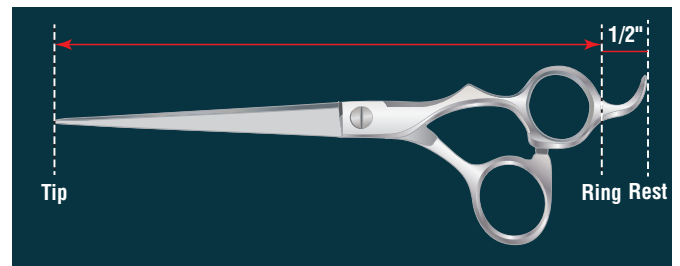


figure 5-16

Opposing handle design.

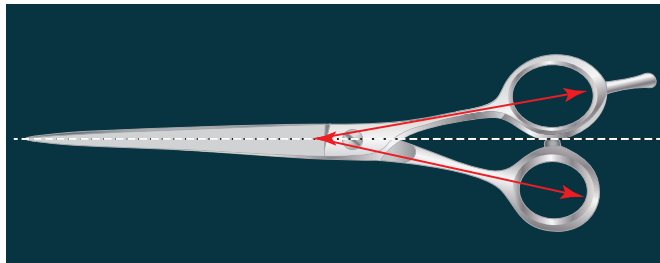
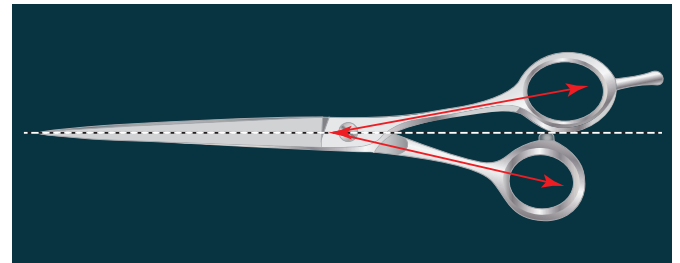


figure 5-17

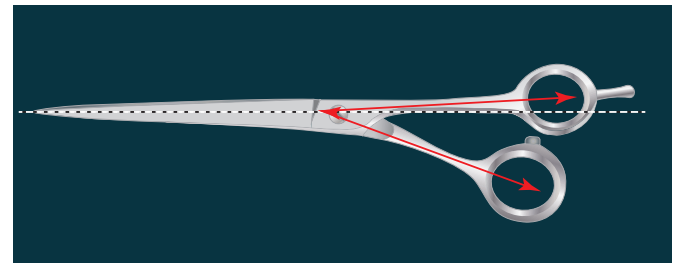
Offset handle design.



- **Handle designs.** Three standard handle designs are the opposing grip, the offset grip, and the crane grip. In the opposing handle design, the finger and thumb grips are positioned directly across from each other (Figure 5-16). The offset handle design has a shorter thumb shank to reduce overextension and is considered to be more ergonomically correct than the opposing design (Figure 5-17). The crane handle also has offset grips but the angle of the finger grip is less in relation to the center of the shears and the thumb grip (Figure 5-18). This reduces pressure and stress in the hand and thumb. Refer to Figures 5-16 to 5-18 to compare handle designs.

figure 5-18

Crane handle design.



DESCRIBE TEXTURIZING, THINNING, AND BLENDING SHEARS

Texturizing (TEKS-churyz-ing), **thinning** (THIN-ing), and blending shears are designed with blades that have teeth of varying widths and distances between them (Figure 5-19). The wider the spacing of the teeth, the more hair is removed during cutting. The greater the number of teeth, the more finely the hair strands can be cut without leaving noticeable cut marks. Although the terms *texturizing shears*, *thinning shears*, and *blending shears* tend to be used interchangeably within the industry, there are some differences in their designs. Refer to Table 5-3 for a summary of these differences.

figure 5-19

Thinning or texturizing shears.



table 5-3

COMPARE TEXTURIZING, THINNING, AND BLENDING SHEARS

Texturizing Shears	Thinning and Blending Shears
Chunking shears: 5 to 9 widely spaced teeth; creates patterns and texture in hair. Texturizing shears: 14 to 28 medium width teeth; differences in hair lengths are visible; adds texture and volume.	Thinning and blending shears: 30 to 50 thin, narrowly spaced teeth that eliminate visible lines in hair; may be designed with teeth on one or both blades; used for blending hair ends and removing bulk or weight.

Show How to Hold the Shears and Comb

After reading this section, you will be able to:

- LO5 Show how to properly hold shears for haircutting.
- LO6 Show how to palm the shears and comb.

figure 5-20
Correct finger placement 1.



figure 5-21
Correct finger placement 2.



figure 5-22
Incorrect finger placement.



Holding your tools properly will give you the most control and best results when cutting hair and helps to avoid muscle strain in your hands, arms, neck, and back. Haircutting shears are designed to be held and manipulated differently than basic household scissors, so the position of your fingers and hand may feel awkward at first. This is to be expected but a little practice will have you being comfortable with it in no time.

During a haircut, you will be holding the comb and shears at the same time throughout the entire cut. Doing this properly requires that you learn how to palm the shears and comb safely. Knowing how to palm the shears and comb will also help you avoid setting the tool down during a cut, which not only wastes time but also requires that the item be disinfected before you can use it again. Review the following sections and practice, practice, practice!

HOLDING THE SHEARS

The following points indicate the procedure for holding the shears.

1. Insert your ring finger into the finger grip (still blade) approximately halfway between the first and second knuckle with the little finger resting on the finger brace. Do not allow the finger grip to slide past the second knuckle or you will lose control of the shears. To ensure proper balance, rest the index and second fingers on the shank of the still blade.
2. Simultaneously, insert the tip of your thumb into the thumb grip (moving blade) no further than your cuticle, which is about halfway between the end of your thumb and the first knuckle. Do not allow the thumb grip to slide below the first knuckle or you will lose control of the cutting blade. See [Figures 5-20](#) and [5-21](#) for the correct finger placement and holding position of the shears. Incorrect finger placement is shown in [Figure 5-22](#).

3. Practice opening and closing the shears, using only the thumb to manipulate the moving blade of the shears. To gain more control during practice, rest the tip of the shears on the first finger of your other hand to steady the still blade while you open and close the moving blade with your thumb.

HOLDING THE SHEARS AND COMB

For safety, shears need to be closed and resting in the palm while you comb the hair; this is called **palming the shears** (PAHM-ing THE SHEERZ). Transferring the comb to the opposite hand after combing to facilitate haircutting is called **palming the comb** (PAHM-ing THE KOM).

1. **Palming the shears.** To palm the shears, slip your thumb out of the thumb grip and simply pivot the shears into the palm of your hand (**Figure 5-23**).
2. **Transferring and palming the comb.** Once the shears have been palmed, hold the comb between the thumb and first two fingers of the same hand to comb through the hair (**Figure 5-24**). After the hair section has been combed, transfer the comb into the palm of the opposite hand between the thumb and the base of the index finger (**Figure 5-25**). This procedure allows the first two fingers of that hand to be free to control the hair while the shears hand is free to cut the hair section (**Figure 5-26**).

RIGHT-HANDED VERSUS LEFT-HANDED SHEARS

There is a difference between right-handed and left-handed shears. Not only are the blade designs different, but the assembly of left-handed shears is reversed during manufacture. Simply switching hands or turning right-handed shears over does not make it a functional tool for left-handed barbers.

MAINTAINING YOUR SHEARS

It is important to maintain your shears on a regular basis to keep them in good working order. Use the following as a guide to protect the investment you have made in your shears.

figure 5-23
Palming shears.



figure 5-24
Holding comb and shears.



figure 5-25
Palming the comb.



figure 5-26
Correct palming of comb while cutting.



- Clean and disinfect the blades after each use according to state board regulations and manufacturer's directions.
- Adjust the tension screw daily or as needed to maintain proper functioning using the following steps:
 1. Hold the shears horizontally by the thumb grip (moving blade) between the thumb and index finger of the left hand with the tension screw facing you.
 2. Raise the finger grip (still blade) to an open position with the thumb and index finger of the right hand. Release the still blade. The blade should drop 25 percent to 35 percent of the way closed before stopping, depending on the quality and style of the shears. If the blades do not drop at all, the tension is too tight and you will feel resistance while opening or closing them. If the blades close completely, they are too loose and will end up bending the hair over the blades, rather than cutting it.
 3. To adjust a slotted tension screw, close the shears and place the blades flat on a hard surface. Hold the shears securely and use the appropriate size screwdriver to turn the screw in gradual increments, checking the tension after each adjustment.
 4. To adjust a knob tension screw, turn the knob to the right one increment at a time and recheck the tension after each adjustment.

Dos and Don'ts for Handling Your Shears:

- Avoid dropping shears. Even one drop on a hard surface can ruin the set or damage the blades.
- Protect shears in a sheath or holder during transport.
- Never cut anything but human hair with haircutting shears! Use a less expensive shear on mannequin hair and save your finest tools for client services.
- Never force shear blades through a section of hair. If there is resistance, section off a thinner parting for cutting.
- Avoid prolonged immersion of shears in corrosive chemicals including permanent waving lotions, oxidizers, and certain disinfectants.
- Gently place shears into disinfectant containers to avoid damaging the tips.



DID YOU KNOW?

The term *shears* is used to describe a haircutting tool that is longer than 6 inches.

The term *scissors* is used when describing a cutting tool measuring less than 6 inches.

Know about Clippers and Outliners

After reading this section, you will be able to:

- LO 7 Describe two types of clippers.
- LO 8 Identify the main parts of a clipper.

Clippers (KLIP-uhz) and outliners are two of the most important tools used in barbering. Clippers can be used for a variety of cutting techniques, from blending to texturizing. Outliners, also known as **trimmers** (TRIM-uhz) or edgers, are essential tools for finish and detail work. Today's barber has a vast array of clipper styles from which to choose. Function,

style, weight, contour, power, and single-speed or two-speed options are just some of the factors that should be considered when purchasing a clipper.

The two main types of clippers are detachable-blade clippers and adjustable-blade clippers. Detachable-blade clippers utilize removable blades in a variety of sizes to achieve different hair lengths. Blade changes are easily accomplished by removing and attaching blades while the clipper is running. Adjustable-blade clippers have adjustable blades that are affixed to the unit with screws. The blades are opened or closed by the adjustment lever on the side of the unit. Clipper guards fit over the blades to extend the length of the hair beyond that left with the blades fully opened, which is about $\frac{1}{8}$ inches depending on the manufacturer. Check online or with your local supplier for the latest clipper models and styles.

KNOW THE PARTS OF A CLIPPER

The visible parts of a clipper vary depending on whether the unit is a detachable-blade clipper or an adjustable-blade clipper and whether it is corded or rechargeable. Some detachable clipper models have separate oil reservoirs and carbon brush cover screws that are accessible at the side of the unit and others do not. At the minimum, the visible parts of a detachable clipper are the detachable still and moving blade assembly, on-off switch, hanger loop, and conducting cord. The visible parts of a typical adjustable-blade clipper are the cutting blade, still blade, heel, on-off switch, blade adjustment lever, set or power screw, hanger loop, and the conducting cord (Figure 5-27). Electric clippers are driven by one of three basic motor types: rotary, pivot, or magnetic (vibratory). Refer to Figures 5-28 to 5-30 and Table 5-4 to compare the features of different types of motors.

figure 5-27
Visible parts of an electric clipper.

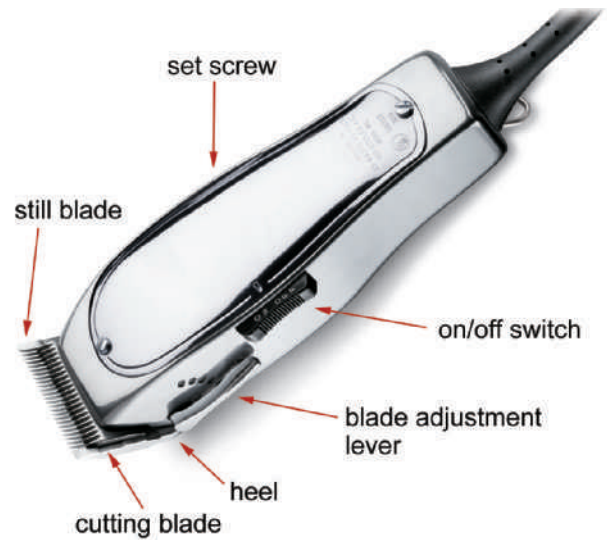


figure 5-28
Rotary (universal) motor clippers.



figure 5-29
Pivot motor clippers.



figure 5-30
Magnetic motor clippers.



table 5-4

COMPARE CLIPPER MOTORS

Type of Motor	Power	Blade Speed	Characteristics	Hair Types
Rotary (universal)	High	High	Heavy-duty use; more frequent replacement of parts	All types, damp or dry
Pivot	High	Lower	Blades pulled in both directions; quiet, cool running	Thick, coarse, or damp
Magnetic	Reliable	High	Alternating spring and magnet mechanism; pulls blades in one direction; long lasting	Tapering and general cutting; dry cutting on fine textures

OUTLINERS

An outliner uses a magnetic or pivot motor. Outliners have very fine cutting blades that require very little pressure against the skin to be effective cutting tools. Using too much pressure can cause skin irritations, abrasions, and even ingrown hairs, so use a light hand and let the tool do the work.

- The cutting blade is usually available in two styles: a straight trimmer blade (Figure 5-31) and a T-shaped blade (Figure 5-32).
- The versatility and utility of the T-blade when trimming rounded or difficult areas makes it the first choice of many barbers and stylists.
- Outliners are essential tools for detail, precision design, and fine finish work on haircuts and facial hair trims, including the eyebrows.

Cordless Clippers and Outliners

A number of manufacturers have produced clippers and outliners that do not have a conducting cord attached directly to the unit. Instead, these tools are designed to rest in a charging unit when not in use. Cordless clippers are an important innovation for barbers because they are easily maneuverable and portable; however, they may not have the same power as corded models.

figure 5-31
Straight-blade trimmer.



figure 5-32
T-shaped blade trimmer.



© Courtesy of the Andis Company.

© Courtesy of the Andis Company.

- Detachable-blade sizes can differ from one manufacturer to another and may not always indicate the same cutting length; so be careful when purchasing these items. Visit manufacturers' websites for photos and size charts.
- A detachable 00000 blade produces the closest cut. Most adjustable-blade clippers have a blade length range from a 000 in the closed position to a 1 in the open position.
- Follow the manufacturer's recommendations for the style and size of clipper blades that are appropriate for their clipper models.
- Manufacturers are constantly improving their clipper blades to permit faster and more precise haircutting, so be on the lookout for the newest in haircutting tools (**Figure 5-33**).

- The purpose of a clipper guard is to allow the hair to be left longer than what might be achieved from using clipper blades alone against the scalp surface.
- Although guards can help to ensure uniformity within the cut, they may also require more passes over a section because they cannot grip the hair the way steel blades do. This can result in lost tension and slippage that may cause steps or gaps in the hair, requiring repeated passes with the clipper. Remember, guards do not do the actual cutting and should not be confused with the clipper blades; they are simply supplemental implements you may choose to use in haircutting services (**Figure 5-34**).

After reading this section, you will be able to:

The technique used by barbers to hold the clipper is most often determined by the section of the head they are working on. For example, cutting the back section will necessitate holding the clippers differently than when



 Check with your instructor about the use of guards on the state board examination.

DID YOU KNOW?

You can visit the following web-sites to learn more about clippers, blades, maintenance, and other related topics:

Andis: <http://www.andis.com>

Oster: <http://www.osterstyle.com>

Wahl: <http://www.wahlpro.com>

DID YOU KNOW?

When using clippers, the amount of hair that remains on the head depends on whether the hair is cut *with* the grain or *against* the grain. Cutting against the grain cuts the hair shorter than cutting with the grain.

CAUTION

Never adjust outliner blades flush to each other. Doing so can create a cutting edge so sharp that it may cut the skin or cause skin irritations and ingrown hairs.

- Blades should only be set as recommended by the manufacturer.
- Typically, the top or front blade should rest between $\frac{1}{16}$ and $\frac{1}{32}$ inches below the back blade, depending on the tool.
- If the client requests a closer cut than what the clipper blade will produce, recommend a head shave.

DID YOU KNOW?

Manufacturer's cleaning and maintenance directions should be followed carefully to ensure optimal performance and to maintain product warranties.

cutting the top section. A general rule to follow is to hold the clippers in a manner that permits free wrist movement while providing the best access to the area being cut. Four holding positions are explained as follows, but be guided by your instructor for alternative methods.

- Place the thumb on top of the clipper with the fingers supporting it from the underside (**Figure 5-35**). This position is usually comfortable for tapering in the nape or side areas of a haircut or when the clipper is switched to the left hand while cutting hair sections from a different direction.
- An alternative method is to place the thumb on the left side of the unit, at the switch, and the fingers on the right side, with the blades pointing up (**Figure 5-36**). Depending on the style and weight of the clippers, you may find that this position limits the range of your wrist action.
- **Figures 5-37** and **5-38** show alternative underhand positions that may be used when working haircut sections from side-view and back-view positions.

figure 5-35
Clipper holding position 1.



figure 5-36
Clipper holding position 2.



figure 5-37
Clipper holding position 3.



figure 5-38
Clipper holding position 4.



Know about Straight Razors

After reading this section, you will be able to:

- LO¹⁰ Name two types of straight razors.
- LO¹¹ Identify the different parts of a straight razor.

As the sharpest and closest cutting tool, razors are used for facial shaves, head shaves, neck shaves, finish work around the sideburn and behind-the-ear areas, and haircutting. The razor of choice for professional barbering is the straight razor; safety razors are not used to perform barbering services in the barbershop.

The two types of straight razors used in barbering are the **changeable-blade straight razor** (CHAYNJE-able BLAYD STRAYT RAY-zor) and the **conventional straight razor** (kun-VEN-shun-ul STRAYT RAY-zor). Both may be purchased with a razor guard for use in razor-cutting the hair (see Chapter 13).

Selecting the right kind of razor is a matter of personal choice. The best guides for buying high-quality razors are as follows:

- Consult with a reliable company representative or salesperson who can recommend the type of razor best suited to your work.
- Consult with more experienced barbers about which razors they have found best for shaving and haircutting.
- Experiment with a variety of razors to determine the style and type most comfortable for you based on the service to be performed. For example, you may prefer a longer blade razor when performing a facial shave and a shorter blade razor for arching around the ears during a haircut.
- Avoid judging a razor simply on color or design. Neither one of these characteristics provides a true indication of the razor's caliber as a cutting tool.

IDENTIFY THE PARTS OF A STRAIGHT RAZOR

The structural parts of conventional and changeable-blade straight razors are basically the same except in the blade area; the conventional razor has a blade and the changeable-blade razor has a blade holder for the blade. Therefore, the structural parts of a straight razor are the head, back, shoulder, tang, shank, heel, edge, point, blade or blade holder, pivot, and handle (Figure 5-39).

DESCRIBE CHANGEABLE-BLADE STRAIGHT RAZORS

The changeable- (or disposable-) blade straight razor closely resembles a conventional straight razor in its overall design. This type of razor tends to be used almost exclusively in the barbershop because its disposable blade eliminates honing and stropping, saves time, and helps to maintain infection control standards (Figure 5-40).

figure 5-39
Parts of a razor.

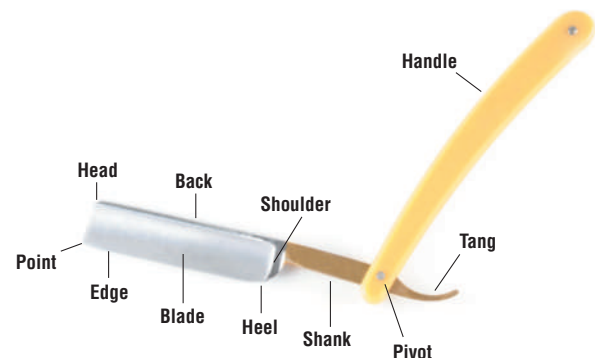


figure 5-40
Changeable-blade razor.



figure 5-41
Razor shaper.



figure 5-42
Hair razor.



Razor Shapers

There are a variety of razor shapers (also known as hair shapers) available on the market today. One style made of lightweight metal with a pivot and open-handle construction leaves the majority of the blade visible (Figure 5-41). This type of razor is not recommended for facial shaving because the blade size and shape can make it challenging to safely shave smaller areas, such as above the upper lip. Other hair shaper designs do not have a pivot and consist of a stationary handle with a finger rest (Figure 5-42). Both styles use a changeable-blade system and come with a guard.

FOLLOW MANUFACTURER'S DIRECTIONS FOR CHANGING THE BLADE

There are several changeable-razor models available and each will have its own design for holding the blade in place; therefore, the method for replacing the blade will vary depending on the model.

Show How to Hold a Straight Razor

After reading this section, you will be able to:

- LO12 Show how to hold a straight razor for shaving, honing, and stropping.
- LO13 Show how to hold a straight razor for haircutting.
- LO14 Describe the functions of hones and strops.
- LO15 Show how to hone and strop a conventional blade straight razor.

figure 5-43
Stroke with razor.



STATE REGULATORY ALERT!

Some state barber boards prohibit the use of conventional straight razors in the barbershop. Be sure to check the rules and regulations in your state before purchase or use.

HOLDING THE STRAIGHT RAZOR

How you hold the straight razor will depend on the procedure to be performed, the type of razor you use, and the positioning that gives you the most control of the razor. Refer to Figures 5-43 to 5-47 and the following section to consider several holding techniques.

- Shaving: The ball of the thumb and first two fingers are positioned on the flat side of the shanks with the handle pivoted up to allow the little finger to rest on the tang. This positions the razor at a better angle while providing more control over the razor (Figure 5-43).
- Honing and stropping: The ball of the thumb and first two fingers are positioned on the flat sides of the shank with the handle in a straight position (Figures 5-44).
- Haircutting: The ball of the thumb supports the razor at the bottom of the shank and the little finger rests on the tang, with the first two or three fingers at the top of the shank (Figure 5-45). Or, the razor handle is in a straightened position with the thumb and first two fingers almost touching at the shank (Figure 5-46).

- **Holding the comb and razor:** To palm the razor for hair combing, simply roll the razor into your hand with the blade facing away from the comb. Hold the comb between the thumb and first two fingers (**Figure 5-47**). Keep a firm grip on the razor to prevent it from slipping and cutting your hand during combing.

LEARN ABOUT CONVENTIONAL STRAIGHT RAZORS

The conventional straight razor requires honing and stropping to maintain its cutting edge. This razor is composed of a hardened steel blade attached to a handle by means of a pivot. The handle may be constructed of hard rubber, plastic, metal, or new polymer materials.

Consider Razor Quality

In order to determine the quality of a razor, you must consider the following factors: razor balance, temper, grind, finish, size, and style.

Razor Balance

Razor balance refers to the weight and length of the blade relative to that of the handle. A straight razor is properly balanced when the weight of the blade and handle are equal. Proper balance of the razor allows for greater ease and safety in handling the razor during shaving. Opening the razor and resting it on the index finger at the pivot will test the balance of the razor. If the head of the razor moves up or down, the razor is not well balanced.

Razor Temper

Tempering the razor refers to a special heat treatment included in the manufacturing process. When a razor is properly tempered, it acquires the degree of hardness required for a good cutting edge. Razors can be purchased with a hard, soft, or medium temper. The barber should select the temper that produces the most satisfactory shaving results. While hard-tempered razors will hold an edge longer, they are difficult to sharpen; conversely, soft-tempered razors are easier to sharpen, but the sharp edge does not last long. For those reasons, many barbers prefer a medium-tempered razor.

Razor Grind

The grind of a razor is the shape of the blade after it has been ground. There are two general types: the concave grind and the wedge grind (**Figure 5-48**).

- *Concave grind:* The concave grind (often referred to as the hollow ground) is available in full concave, one-half concave, and one-quarter concave forms. The back and edge of the razor looks hollow, being slightly thicker between the hollow part and the extreme edge. Many barbers prefer the hollow-ground razor since the resistance of a beard can be felt more easily and alerts the barber to check the sharpness of the cutting edge. Although the one-half and one-quarter concave grinds are less hollow than the full concave, the outside dimensions of the blade appear the same.
- *Wedge grind:* The wedge grind is neither hollow nor concave. Both sides of the blade form a sharp angle at the extreme edge of the razor. Most older razors were made with a wedge grind. Although learning how to sharpen a wedge grind may be a challenge, once mastered, this grind produces an excellent shave. It is especially preferred for men with coarse, heavy beards.

figure 5-44
Razor strop.



figure 5-45
Holding the razor shaper properly.



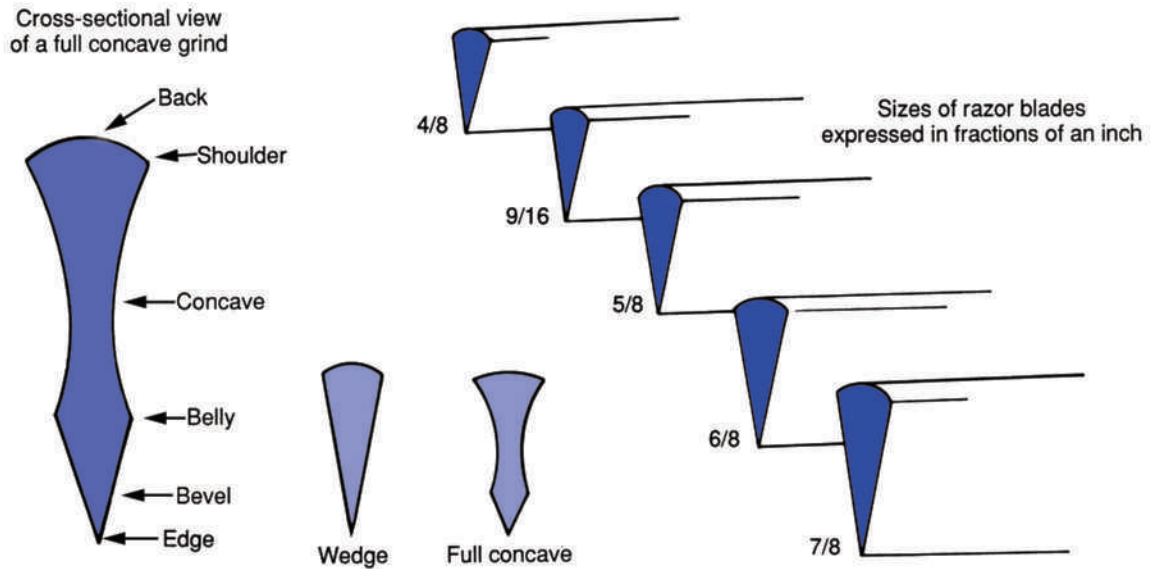
figure 5-46
Alternate method of holding the razor shaper.



figure 5-47
Palming the razor and comb.



figure 5-48
Razor grinds.



Razor Finish

The finish of a razor is the polish of its surface. This finish may be plain, crocus (polished steel), or plated with nickel or silver. Of these types, the crocus finish is usually the choice of the discriminating barber. Although the crocus finish is more costly, it lasts longer and does not rust as easily as other finishes. Metal-plated razors are undesirable because the finish wears off quickly and may hide poor-quality steel.

Razor Size

The size of the razor is measured by the length and width of the blade. The width of the razor is measured either in eighths or sixteenths of an inch, such as $\frac{4}{8}$ inches, $\frac{5}{8}$ inches, $\frac{6}{8}$ inches, $\frac{7}{8}$ inches, and $\frac{9}{16}$ inches. Two of the most common sizes are the $\frac{5}{8}$ inches and $\frac{9}{16}$ inches, the $\frac{5}{8}$ inches being the more popular.

Razor Style

The style of a razor indicates its shape and design. Modern razors have such features as a back and edge that are straight and parallel to each other; a round heel; a square point; and a flat or slightly round handle. To avoid scratching the skin, the barber usually rounds off the square point of the razor slightly by drawing the point of the razor along the edge of the hone.

Razor Care

Razors will maintain their quality if care is taken to prevent corrosion of the extremely fine edge. After use, a razor should be cleaned, stropped, and a little oil applied to the cutting edge. Be careful not to drop the razor as doing so may damage the blade. When closing the razor, be careful that the cutting edge does not strike the handle. If the cutting edge strikes the handle when closing the razor, it may indicate that the handle is warped or that the pivot is too tightly riveted. The barber's tool kit should include several high-grade razors so that a damaged razor can be replaced immediately.

IDENTIFY ACCESSORIES USED WITH CONVENTIONAL STRAIGHT RAZORS

There are two vital accessories used with conventional straight razors: the hone and the strop. A **hone** (HOHN) is an abrasive material that has the ability to cut steel. It is used to grind the steel and impart an effective cutting edge to the razor's blade. A **strop** is a leather and canvas accessory that is used to smooth and align the cutting teeth of the razor edge and polish the blade. Honing and stropping techniques must be mastered to prepare a conventional straight razor for shaving.

HONES

There are various types of hones available for the purpose of sharpening razors. Hones are manufactured in a rectangular block shape, without or with an attached handle, for ease of blade placement on the surface. Since the abrasive material of the hone is harder than steel, it will cut or file the edge on the blade of the razor.

Students should practice with a slow-cutting hone, while experienced practitioners generally use a faster-cutting hone.

There are three main types of hones: natural, synthetic, and combination.

- Natural hones are cut from natural rock formations. The two types of natural hones are the water hone and the Belgium hone. These hones have a slow cutting action that can produce a fine, long-lasting edge when used with water or shaving lather to lubricate the stone before sharpening.
- Synthetic hones, such as the carborundum hone, are manufactured products. These hones cut faster than water hones, producing a keen cutting edge in less time and may be used wet or dry, although you need to take care not to over-hone the razor.
- Combination hones consist of both a water hone and a synthetic hone. The synthetic side is used first to develop a good cutting edge and the natural side is used to produce a fine finished edge.

Choosing a Hone

When selecting a hone, remember that the finer the abrasive, the slower its action. Many barbers use combination hones; however, it is advisable to be familiar with the other types of hones and to understand the benefits of each. The type of steel in the razor also makes some difference as to whether a good edge can be obtained with a particular type of hone. Be guided by your instructor, personal experimentation with different hones, and razor manufacturer's recommendations.

Care of Hones

Always clean the hone before use. Use water and a pumice stone to remove the tiny steel particles that accumulate on the surface of the hone. If a new hone is very rough, the same method can be used to work it into shape.

When wet honing is done, always wipe the hone dry after use. This aids the cleaning process and also wipes away the particles of steel that adhere to the cutting surface. Always disinfect the hone according to the manufacturer's directions.

Honing Guidelines

Review the following guidelines before practicing the honing techniques in Procedure 5-1.

- Position the index finger along the top of the shank to ensure even pressure against the blade while honing.
- Make sure your fingertips do not project above the edge of the hone to avoid injury.
- The razor is stroked *edge-first* diagonally across the hone. This produces teeth with a cutting edge.
- The edge blade should be kept flat on the hone, with no rocking or lifting from the surface.
- Stroke the blade with equal pressure from heel to point and from side to side.
- An equal number of strokes should be made on both sides of the blade.
- The angle at which the blade is stroked must be the same for both of its sides.
- As the blade edge is sharpened, gradually lighten the pressure and test frequently.
- The number of strokes required in honing depends on the condition of the razor's edge.

figure 5-49
Testing a honed razor.



CAUTION

This testing technique requires a great deal of practice and experience. Be very cautious of the amount of pressure and speed with which the test is performed. Touch the razor's edge lightly, and note the sensation you feel. A dull edge does not produce a drawing sensation. A proper cutting edge will have a sharp drawing sensation. If the razor's edge yields a smooth feeling upon testing, finish it again on the canvas strop, followed by a few more strokes on the leather strop.

Testing a Honed Blade

A honed blade is tested by lightly passing it over a thumbnail moistened with water or lather ([Figure 5-49](#)). You should feel one of the following sensations as the blade edge passes over your nail:

- A keen edge has fine teeth and tends to dig into the nail with a smooth, steady grip.
- A blunt or dull razor edge passes over the nail smoothly without any cutting power.
- A coarse razor edge digs into the nail with a jerky feeling.
- A coarse or over-honed edge has large teeth that stick to the nail and produce a harsh, grating sound.
- A nick in the razor produces the feeling of a slight gap or unevenness when drawn across the nail.

5-1 Honing the Razor *pages 133–134*

STROPS

Unlike hones, which are designed to grind the edge of a razor into a sharp cutting edge, strops are used to remove small metal particles from the blade,

to smooth the edge, and to polish the razor. A good strop is made of durable and flexible material with the proper thickness and texture and shows a smooth, finished surface. Some barbers like a thin strop; others prefer a thick, heavy strop.

Strops are generally classified as leather or canvas. Cowhide and horsehide are the most common leathers used to manufacture a strop. The four types of strops covered in this chapter are the combination, canvas, Russian, and shell strops.

- The combination strop is the type of strop most commonly used in the barbershop because it is an all-in-one accessory for preparing the razor after honing. This strop is actually made of two strops—a canvas strop and a leather strop—that have been riveted together at the attachment end of the strop.
- A **canvas strop** (can-viss strohp), or the canvas side of a combination strop, should be made of high-quality linen or silk, woven into a fine or coarse texture. A fine-textured canvas strop is desirable for removing any metal burrs or imbrications that remain after honing. To obtain the best results, a new canvas strop should be thoroughly broken in. Consult manufacturer's instructions for recommended products and directions to prepare and maintain a canvas strop.
- The **Russian strop** (RUSH-an strohp) was originally imported from Russia and still bears the name even though it may be manufactured elsewhere. The name simply implies that the Russian method of tanning was used to prepare the leather. Russian strops are usually made from cowhide and are considered to be one of the best. Consult the manufacturer's instructions for recommended products and directions to prepare and maintain a Russian strop.
- A **shell strop** is a high-quality strop taken from the muscular rump area of a horse. Although shell strops can be expensive, they are considered to be one of the best strops for barbers as they tend to remain smooth and require very little, if any, breaking in. As with other strops, consult the manufacturer's instructions for recommended products and directions to prepare and maintain your shell strop.

Strop Dressing

Strop dressing cleans the leather strop, preserves its finish, and also improves its draw and sharpening qualities. Refer to the manufacturer's directions for proper application methods, timing in between applications, and follow-up procedures.

STROPPING THE RAZOR

From a technical standpoint, the razor may be stropped from the barber's hand toward the chair or from the chair to the hand. When performed correctly, both methods can achieve the same results; however, your state barber board may have a preferred method. Be sure to check the rules, regulations, and exam performance requirements in your state.



DID YOU KNOW?

Finer-quality strops are usually "broken in" by the manufacturer, thereby requiring less breaking in and preparation by the barber.



DID YOU KNOW?

If a strop is labeled "Russian shell strop," it indicates that the strop is made from the rump area of a horse and that the Russian tanning method was used in its manufacture.



DID YOU KNOW?

The direction of the razor's edge used in stropping is the reverse of the direction used in honing.

Stropping Guidelines

Review the following guidelines before practicing the stropping techniques in Procedure 5-2.

- The strop is attached to the arm of the barber chair by a closed clip.
- Hold the end of the strop firmly in your nondominant hand so it cannot sag and on a slight diagonal from the chair at a comfortable height.
- When holding the razor, the index finger is on the shank, the subsequent fingers are on the handle, and the thumb rests at the pivot. The index finger of your dominant hand should rest along the edge of the strop.
- The direction of the blade edge in stropping is the reverse of that used in honing; therefore, the back of the razor will lead each stroke, rather than the blade edge.
- Stropping the razor requires being able to roll the razor on its back after completing a stroke to position it for the next one. With the thumb on the side of the shank closest to you and the first two fingers braced on the other side of the shank, use your thumb to roll the blade up and over for the next stroke.
- Strokes should be made in a single, slightly diagonal stroke against the strop, with even pressure from the heel to the point of the razor.
- The blade edge needs to be flat against the surface to avoid cutting or nicking the strop.
- Bear down just enough to feel the razor draw against the strop.
- Do not worry about speed; a moderate pace is preferred.
- Strokes should be repeated as necessary to finish the razor edge.

CAUTION

When testing the razor's edge on the moistened tip of the thumb, a proper cutting edge will have a sharp drawing sensation. Use minimal pressure and speed to perform the test safely.

5-2 Stropping the Razor *page 135*

Learn about Equipment and Supplies Used in Barbering

After reading this section, you will be able to:

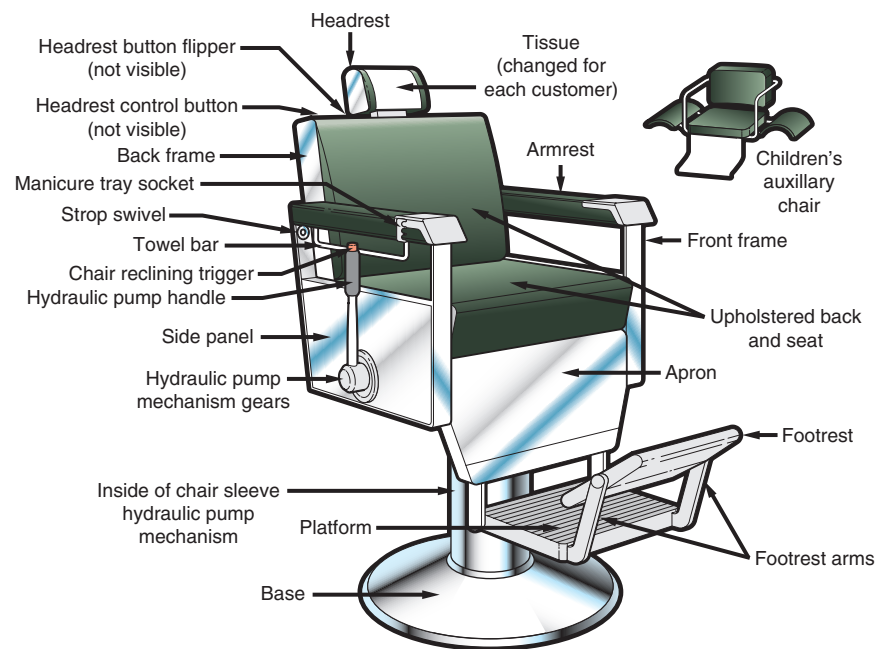
16 Identify the types of equipment and supplies used in barbering.

In addition to your tools and implements, you will need certain equipment and supplies to perform barbering services. Review the following descriptions to become familiar with some common items used in the barbershop.

KNOW THE FEATURES OF A BARBER CHAIR

The barber chair (**Figure 5-50**) is an essential piece of equipment for performing barbering services and should not be confused with a *styling* chair. Styling chairs are typically smaller in design and do not usually have a headrest. Conversely, barber chairs are larger in size, have a headrest, either

figure 5-50
A hydraulic barber chair.



removable or built-in, and are designed to enable clients to fully recline when receiving shaving and facial services.

Barber chairs are available with either hydraulic or motorized bases; both types provide easy height adjustment of the chair. The circumference of the base should be large and heavy enough to accommodate a man's weight without tipping when the customer is in a fully reclined position.

IDENTIFY TYPES OF DRAPES AND CAPES

The two main types of drapes used are shampoo capes and haircutting capes (also known as chair cloths).

- *Shampoo capes* are typically waterproof drapes made of vinyl used to protect the client's skin and clothing from water, liquids, and chemical processes.
- *Haircutting capes* are made of nylon or other synthetic materials. These draping fabrics are usually more comfortable for the client because they do not hold in as much body heat as vinyl capes.

MAINTAIN A SUFFICIENT SUPPLY OF TOWELS AND LINENS

The two types of towels generally used in the barbershop are 100 percent cotton cloth towels and disposable paper towels, especially designed for the profession.

- Cotton towels are available in different weaves and textures. Terry cloth-type towels may be preferred for shampooing and chemical services because of their absorbent qualities; these towels also hold heat and moisture well for steaming the face or scalp.

- Some professionals prefer the type of cotton towel that is traditionally known as a barber towel. This towel has a tighter, flatter weave that is an excellent choice when terry cloth towels are too bulky.
- Disposable paper towels are extra large and measure at least 12 × 24 inches. These multipurpose towels are soft, lint-free, and commonly used as a
 - barrier at the neck instead of a neck strip;
 - wiping paper during shaves;
 - towel wrap for product application or hair removal purposes;
 - sanitary surface on work stations for tools or implements;
 - drawer liner.

In addition to using towels for services or draping purposes, you should know the correct way to towel-wrap your hand to perform the following activities:

- Cleansing and drying the face
- Applying products to the face
- Removing all traces of product, lather, or loose hair from the face, neck, ear areas, and forehead

Practice Procedure 5-3 to master the towel-wrapping methods used when performing barbering services.

- Neck strips are flexible, disposable paper strips used to provide a barrier between the neckband of the cape and the client's neck. Neck strips and hair wrapping strips are not the same thing! Good quality neck strips should be soft and give a little when stretched; hair wrapping strips are usually wider and made of stiffer material that may scratch the skin. See Chapter 11 for the correct way to apply and secure a neck strip.



REMINDER

If permitted in your state, neck dusters must be cleaned and disinfected before each use on the skin.

P 5-3 Towel Wrapping *pages 136–137*

DESCRIBE THE APPLIANCES USED IN BARBERING SERVICES

Appliances used in the barbershop typically include hot towel cabinets, electric latherizers, blowdryers, massagers, and thermal styling tools. Other appliances, such as high frequency and galvanic machines, may be used to provide advanced skin and scalp care services if permissible in your state.

Hot Towel Cabinet

The hot towel or steam towel cabinet is the ideal appliance for maintaining a ready supply of warm towels for barbering services. Warm towels are used during facials and shaves, but they are also appreciated during a manicure, scalp treatment, or after a neck shave (**Figure 5-51**).

Latherizers

Lather receptacles are containers used to hold, heat, or dispense lather for shaving. The most basic and commonly used types are the electric latherizer, the press-button can latherizer, and the lather mug with paper lining.

An *electric latherizer* is a lather-making device that is commonly seen in barbershops. This appliance is sanitary, convenient, easy to operate, and

figure 5-51
Hot towel cabinet.



produces rich, warm lather for shaving services (**Figure 5-52**). Follow the manufacturer's directions for proper use and care.

Press-button can latherizers are convenient and sanitary, but they are not as professional as electric latherizers and do not provide warm lather.

Lather mugs are lather receptacles made out of glass, pottery, rubber, or metal (**Figure 5-53**). When the lather mug is used, shaving soap and warm water are mixed thoroughly with the aid of a lather brush. Since the lather mug can harbor bacteria, it requires thorough cleaning and disinfection before and after each use. Disposable paper liners for the shaving mug are also available and help to ensure a sanitary environment. Check your state barber board rules and regulations regarding the use of lather mugs and brushes.

Lather brushes are used to apply the soap lather used to soften the beard. Most barbers favor the number 3 size of lather brush. The vulcanized type is the most durable, since its bristles will not fall apart in hot water. To guard against contaminated brushes, many states have laws requiring that brushes made from animal hair be free of anthrax germs at the time of purchase. These brushes must contain the imprint "Sterilized." Lather brushes must be cleaned and disinfected after each use. *Shaving soaps* can be purchased in various forms and shapes. These soaps usually contain animal and vegetable oils, alkaline substances, and water. The presence of coconut oil improves the lathering qualities of the shaving soap, which helps to keep the hair erect while the alkalinity of the soap softens the hair for shaving.

- Hard shaving soaps include those sold in cake, stick, or powdered form.
- Soft soap is available as shaving cream in a tube, jar, or press-button container.
- Liquid cream soap is used in electric latherizers.

Blowdryers

A blowdryer is an electrical appliance designed for drying and styling hair in a single operation. Standard parts include a handle, directional nozzle (concentrator), small fan, heating element, and speed/heat controls (**Figure 5-54**). Comb, brush, and diffuser attachments are also available with some models.

To keep your blowdryer in safe working order, make sure the unit and air intake at the back are kept clean at all times. A buildup of dirt, hair, or oils can cause the elements to burn out prematurely.

Since there are many different models available, try out several dryers before buying one. You will want to be sure the dryer fits your hand comfortably with the right balance and weight for easy use. Then, review the warranty and service agreements to protect your investment.

The following *blowdryer safety* precautions should be observed:

- Apply a protective blowdrying agent to the hair to shield it from the heat of the dryer.
- Check the comb teeth and brush bristles on attachments for sharp points.
- Test the temperature controls.

figure 5-52
Latherizer.



figure 5-53
Lather mug.



**CAUTION**

Check your state barber board rules and regulations concerning the use and disinfection of lather mugs and brushes.

figure 5-54
Blowdryer.



- Keep the air and the hair moving to prevent burns. Do not concentrate dryer heat too close to the scalp.
- Follow the styling brush or comb with the blowdryer, working in the direction of the desired style, blowing damp air onto dry hair.
- Avoid dropping the dryer.
- Never use the dryer near water.

figure 5-55

Electric massager.



figure 5-56

The parts of a thermal iron.

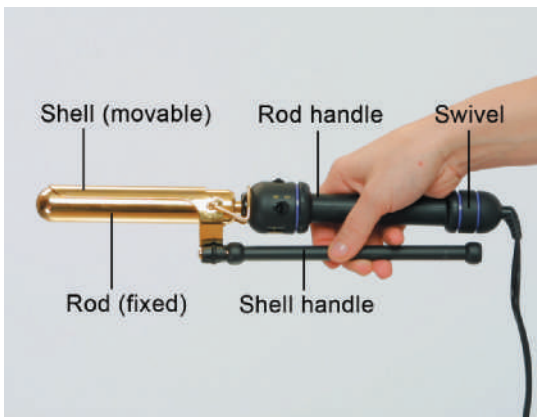


figure 5-57

Flat iron.



Electric Massager

The massager (vibrator) is an electric appliance used in facial, scalp, and shoulder massage services. This device generates vibrations that travel through the barber's fingers to the client's scalp, face, or neck. Some massager units are heavy and may be uncomfortable or difficult for smaller hands to control. Always try one on for size before purchasing (Figure 5-55).

Thermal Styling Tools

Thermal styling tools (THUR-mul STYL-ing TOOLS) create heat that is used to wave, curl, or press hair into another shape or form. Any form of styling performed with a heated tool or appliance may be considered a form of thermal styling. This includes the use of Marcel irons, stoves, hot rollers, flat irons, and other heated tools. Although thermal hairstyling tools are used most often in women's hairstyling, the ability to use these tools correctly is an important aspect of the barber's training. Review Procedure 5-4 on manipulating thermal irons.

- **Thermal irons.** Thermal irons are available in a variety of styles, sizes, and weights, with small to large barrel diameters, and may be electric or require heating on a stove. The four basic parts of thermal irons are the rod handle, shell handle, barrel, and shell (Figure 5-56).
- Electric curling irons are available with or without a vaporizing action feature and can be purchased with a stationary or rotating handle.
- Electric flat irons are used to temporarily straighten curly or wavy hair. They can also be used to give direction to straighter hair textures while imparting a glossy, finished look to the hair. Flat irons are available in many sizes, from mini-irons for short hair lengths or tight styling areas to larger irons suitable for long-hair styling (Figure 5-57).
- The *conventional (Marcel) iron* requires the use of a stove to heat it (Figure 5-58). *Pressing combs* also require the use of a stove and are used for straightening or pressing hair. Electric pressing combs are also available. Both types of combs are constructed of high-quality steel or brass.
- The *electric stove* is used to heat conventional irons and metal pressing combs. Stoves become very hot and must be handled carefully (Figure 5-59).